Project Title

Automating Employee Data Using ServiceNow

2. Objective

The main objective of this project is to design and implement an automated system that can manage employee data effectively using the ServiceNow platform. In many organizations, employee information is maintained manually, which can lead to errors, delays, and difficulties in managing large amounts of data. This project aims to eliminate these problems by introducing automation into the employee data management process.

The system allows HR staff or administrators to enter employee information through a digital form. Once the information is entered, it is automatically stored in a ServiceNow database table. Automation workflows then process the data and perform necessary actions such as updating records or sending notifications.

Another important objective of the project is to improve efficiency and reduce manual effort involved in managing employee records. The automated system ensures that employee data is stored accurately, organized properly, and can be easily accessed whenever required. Overall, the project demonstrates how modern platforms like ServiceNow can improve the management of organizational data.

3. Tools / Platform Used

The following technologies were used to develop the system:

- ServiceNow Developer Instance
- ServiceNow Tables (Database)
- ServiceNow Forms
- Workflow Automation / Flow Designer
- Notifications
- Import Sets (for data loading)

The ServiceNow Developer Instance was used as the main development environment for creating the application. It provides a workspace where developers can design tables, forms, workflows, and automation processes.

ServiceNow Tables

Tables in ServiceNow act as a database for storing information. In this project, a table was created to store employee data such as employee name, department, designation, email, and other relevant information.

ServiceNow Forms

Forms were used to collect employee details from users. HR staff or administrators can enter employee data through these forms. Once the form is submitted, the data is automatically stored in the ServiceNow database table.

Workflow Automation / Flow Designer

Workflow automation or Flow Designer was used to automate the processing of employee data. It allows the system to perform actions automatically without manual intervention.

Notifications

Notifications were implemented to inform administrators or users whenever employee data is created or updated.

Together, these tools helped in developing a fully functional automated system for managing employee information efficiently.

4.Methodology

Step 1 – Create Employee Data Table

In this step, a ServiceNow table was created to store and manage employee information in a structured manner. The table acts as the main database where all employee records are stored and maintained. Each record in the table represents the details of a specific employee working in the organization. Creating this table ensures that employee data can be stored securely, organized properly, and accessed easily whenever required.

The table contains several important fields that capture employee information. Some of the key fields included in the table are:

• Employee Name
• Employee ID
• Department
• Designation
• Email
• Phone Number
• Manager
• Date of Joining

• Work Location
• Employment Status

These fields help store complete employee information within the system. By organizing the data in a table format, the system allows administrators to easily view, update, and manage employee records.

The creation of the employee data table is an important step because it provides the foundation for the entire automated system. Once the table is created, other components such as forms, workflows, and notifications can interact with this table to process and manage employee data efficiently.

Step 2 – Create Employee Data Form

In this step, a user-friendly form interface was created in the ServiceNow platform to allow HR staff or administrators to enter and manage employee information easily. The form acts as the main interface through which users interact with the system. It is designed in a structured format so that users can fill in the required employee details without confusion.

The form contains multiple fields corresponding to the employee data table, such as employee name, department, designation, email, manager, and other important details. When users enter the required information and submit the form, the data is automatically stored in the ServiceNow database table.

The form allows users to perform several important actions, including:

• Add new employee records
• Update existing employee information
• View stored employee data
• Search for specific employee records
• Edit employee details when changes occur
• Delete employee records if necessary
• Validate entered data to reduce errors

The form is designed to be simple, organized, and easy to use so that users can quickly enter employee details without technical difficulty. By providing a structured interface for data entry, the form helps ensure that employee information is collected accurately and stored properly in the system.

This step is important because the form serves as the primary interface between users and the automated employee data management system.

Step 3 – Store Data in ServiceNow Database

In this step, the employee information entered by the user through the ServiceNow form is automatically stored in the ServiceNow database table. Once the form is submitted, the system processes the input data and saves it in the corresponding fields of the employee data table. This automated storage process ensures that all employee records are maintained digitally and systematically.

The ServiceNow database acts as a centralized storage system where all employee information is kept securely. Each record in the table represents an individual employee and contains all relevant details such as name, department, designation, email, and manager. Storing the data in the database makes it easier for administrators to manage and retrieve employee information whenever required.

This process ensures several important benefits, including:

• Organized storage of employee records
• Easy access to employee information
• Secure storage of employee data
• Quick retrieval of records when needed
• Reduction of manual record keeping
• Improved accuracy and consistency of employee data
• Ability to update or modify records easily
• Better management of large amounts of employee information

By storing employee data in the ServiceNow database, the system ensures that all information is structured, reliable, and readily available for administrative use. This step plays a crucial role in building an efficient automated employee data management system.

Step 4 – Workflow Automation

In this step, automation workflows were created in the ServiceNow platform to process employee data automatically. Workflow automation helps the system perform certain actions without manual intervention whenever employee data is created, updated, or modified. These workflows ensure that the system operates efficiently and reduces the need for repetitive manual tasks.

The workflows are designed to monitor changes in the employee data table. When a user submits or updates employee information through the form, the workflow is automatically triggered. The system then processes the information and performs the required actions according to predefined rules.

The workflow performs several important functions such as:

• Processing employee information
• Updating records in the database
• Sending notifications to administrators
• Automatically validating entered data
• Triggering actions when new employee records are created
• Managing updates when employee details are modified

• Maintaining consistency of employee information
• Reducing the chances of human error

By implementing workflow automation, the system ensures that employee data management becomes faster, more accurate, and more efficient. It minimizes manual effort and allows administrators to focus on more important tasks instead of handling repetitive data processing activities.

Step 5 – Notifications

In this step, the notification feature was implemented in the ServiceNow system to keep administrators and relevant users informed about important updates related to employee data. Notifications are automatically generated whenever certain actions occur in the system, such as adding new employee records or updating existing employee information. These notifications help ensure that administrators are aware of changes and can monitor employee data management activities effectively.

Notifications can be sent through the ServiceNow platform or via email depending on the configuration. This helps ensure that important information reaches the concerned users quickly and efficiently.

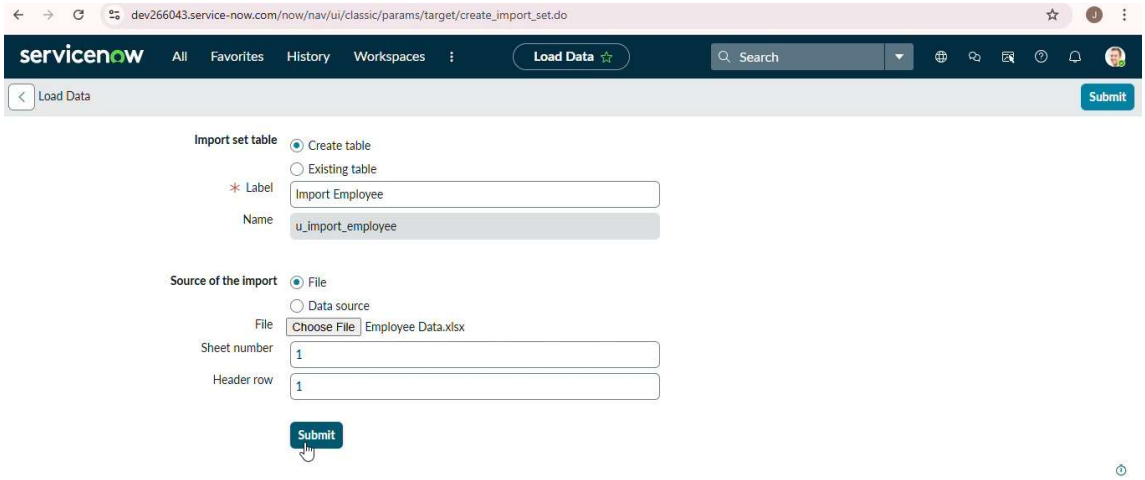
The system generates notifications whenever:

• New employee data is added
• Employee records are updated
• Changes are made to important employee details
• New records are successfully stored in the database
• Administrators need to review updated information
• System activities related to employee management occur

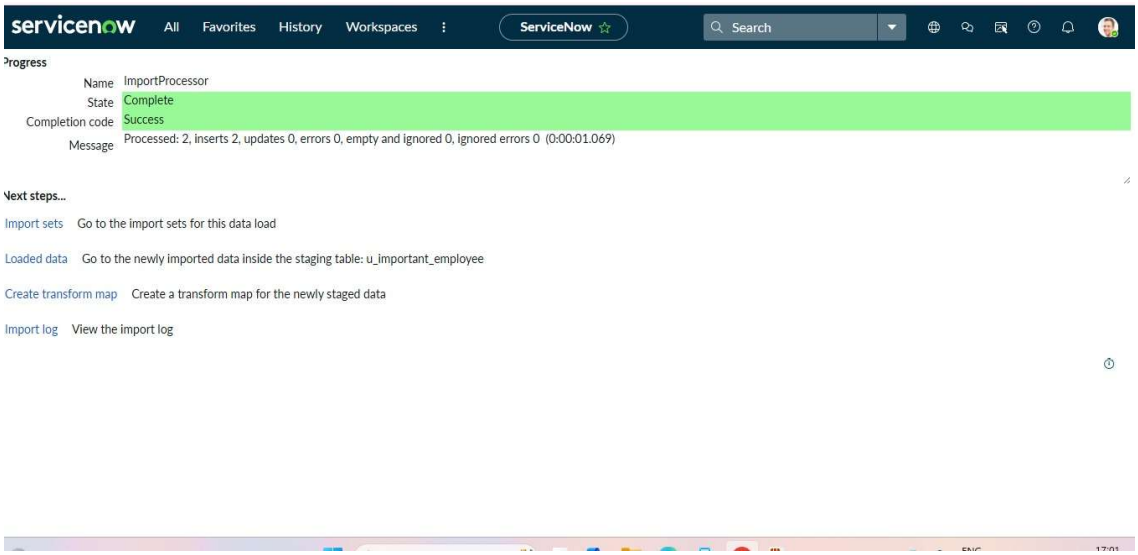
These notifications help administrators stay informed about changes in employee information and ensure better communication within the system. It also helps maintain transparency and allows quick action if any corrections or updates are required.

5.Screenshots

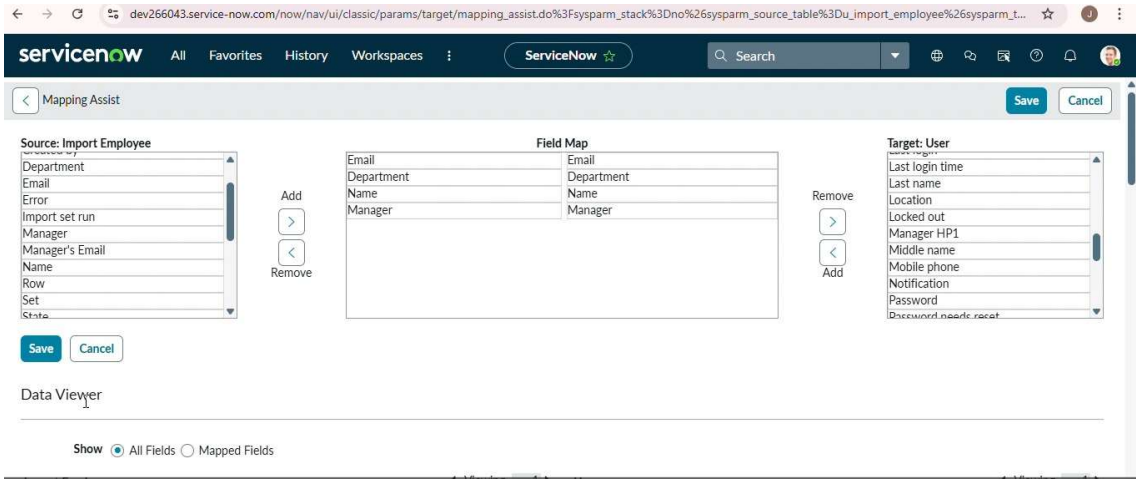
Import Set creation screen



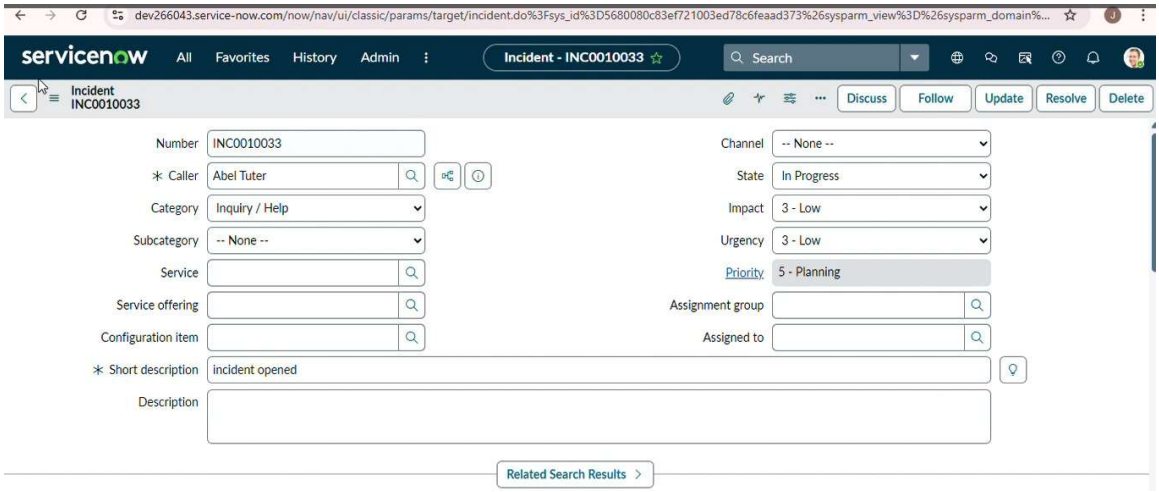
Data loading interface



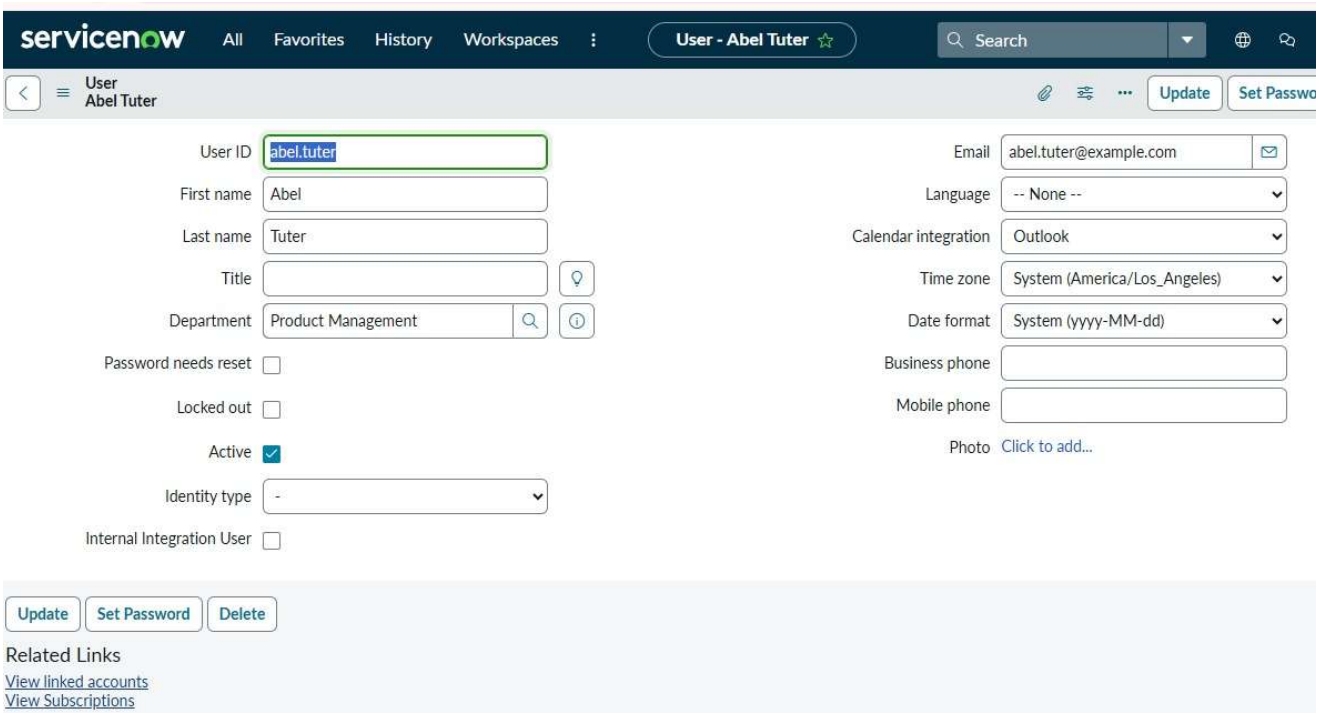
ServiceNow form layout configuration



Incident or employee data entry form



Workflow automation interface



6.Result

After implementing the automated employee data management system, the project produced successful results. The system allows HR staff to enter employee details through a ServiceNow form, and the information is automatically stored in the database. The workflow automation processes the data and performs required actions without manual intervention. Notifications are generated whenever employee records are created or updated, keeping administrators informed. The system improves efficiency by reducing manual work and minimizing the chances of errors in employee data management. It also provides quick access to employee records and ensures that data is organized and maintained properly.

Overall, the project successfully demonstrates how automation can improve the management of employee information within an organization.

7.Conclusion

The Automating Employee Data system demonstrates how modern automation platforms like ServiceNow can simplify and improve employee data management in organizations. By replacing manual processes with automated workflows, the system reduces human effort, improves data accuracy, and enhances overall efficiency.

The project successfully shows how employee information can be collected through forms, stored in database tables, and processed automatically using workflow automation. This approach not only saves time but also ensures that employee records are organized and easily accessible.

In conclusion, the project highlights the importance of automation in modern organizations and demonstrates how ServiceNow can be used to build efficient and reliable systems for managing employee data.